

N200 in the flanker task as a neurobehavioral tool for investigating executive control

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Abstract

Event-related potentials were recorded in a flanker task using arrowheads pointing to the left or to the right as targets and as congruent or incongruent flanker stimuli using squares as neutral flanker stimuli. The onset of the flanker stimuli preceded that of the target stimuli by 100 ms. Lateralized readiness potentials showed response activation below execution threshold in correspondence to the information conveyed by the flanker stimuli. Exclusively, the incongruent flanker condition provoked a N2c, which evolved closely synchronized to the erroneous response. Graded response analyses separating incongruent trials with weak, medium, and strong incorrect response activation revealed that the N2c amplitude covaried with the magnitude of the erroneous response. The N2c in the incongruent compatibility condition of the flanker task thus corresponds to the avoidance of inappropriate responses, possibly reflecting the inhibition of automatically but erroneously primed responses. The results are compatible with studies of error correction, suggesting that efference monitoring is a constituent of executive control.

Descriptors: N200, LRP, Flanker task, Error correction, Response inhibition, Executive control

Two key elements in human information processing are the selection of relevant environmental events and/or the attenuation of irrelevant events and the selection of appropriate actions given current goals and environmental events. During the last few decades, selection of sensory information has been studied intensively. In attention research, the bias toward sensory selection has been criticized repeatedly because the selection of action has been relatively neglected (e.g., Allport, 1989; Neumann, 1990). A decade ago, Logan and Cowan (1984) argued that the regulation of behavior by means of the inhibition of thought and action is a basic act of executive control. To examine this aspect of executive control, we studied response activation and inhibition in the flanker task.

Response Priming

Throughout this paper, *response priming* refers to the activation of responses by partial analysis of stimulus information.

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A paradigmatic choice response time (RT) task that demonstrated response priming was introduced by Eriksen and Eriksen (1974). In this task, a centrally presented target letter was flanked by distractor letters. The target letter was flanked by congruent letters (indicating the same response as the target), by neutral letters (indicating no response assignment), or by incongruent distractor letters (indicating the response opposite to the target). RTs in congruent trials were faster than those in neutral trials, and RTs in incongruent trials were slower than those in neutral trials. Facilitation by the congruent flanker stimuli was attributed to the redundancy between flanker and target information. The incongruent flanker stimuli induced interference between the simultaneously primed, but incompatible, flanker- and target-triggered responses.

According to the continuous flow model (Eriksen & Schultz, 1979), the activation of responses begins as soon as visual information starts accumulating. Early in the process, the accumulated information is consistent with a wide range of responses, and all responses receive initial activation. Because more flanker than target stimuli were presented, flanker-triggered responses received a relatively stronger initial activation than those received by target-triggered responses. Thus, with incongruent flanker and target stimuli, mutually incompatible responses are activated; that is, a response conflict is induced. As information continues to accumulate, response priming becomes increasingly focused on responses that remain viable alternatives given the accumulated data. With the target becoming gradually localized, information from this part of the stimulus pattern gains importance over information derived from the flanker stimuli. In the continuous flow model, the initially primed incorrect responses reduce the readiness to execute the correct response and thus

delay RTs. The speed with which a response is executed depends in part on the extent of response competition. "The greater this competition, the longer the latency of the correct response" (Coles, Gratton, Bashore, Eriksen, & Donchin, 1985, p. 530).

Behavioral and physiological evidence indicated that response priming took place in the flanker task. Gratton, Coles, Sirevaag, Eriksen, and Donchin (1988) proved the existence of response priming by means of the lateralized readiness potential (LRP). The readiness potential over the contralateral motor cortex is more negative than that over the ipsilateral motor cortex in choice RT tasks when a choice is assigned to the responding hand (Kutas & Donchin, 1980). The lateralized part of the readiness potential indicates the preparation of motor responses and gives an index of the relative strength of response priming when responses are assigned to different hands (Coles, 1989). Using the flanker task, Gratton et al. (1988) found that during an early stage of processing the readiness potential over the ipsilateral motor cortex was more negative than that over the contralateral motor cortex in the incongruent compatibility condition of the flanker task. Thus, incorrect responses were more strongly primed during this phase than were correct responses. Analyses of electromyographic (EMG) activity indicated that incorrect responses were primed up to the level of partial execution in a substantial portion of trials in the incongruent flanker condition (Coles et al., 1985; Smid, Mulder, & Mulder, 1990). Similarly, St. James (1990) found that error corrections occurred most frequently in the incongruent flanker condition.

Response Inhibition

Throughout this paper, the term *response inhibition* signifies the interruption of ongoing thought and action by executive control. During the last few years, response inhibition has been studied electrophysiologically. When go and no-go trials in RT tasks are compared, the amplitude of the N200 component of the event-related potential (ERP) has been reported to be enhanced in no-go trials (e.g., Kok, 1986; Pfefferbaum & Ford, 1988), even when go and no-go trials had equal probabilities (Eimer, 1993; Jodo & Kayama, 1992; Schröger, 1993). Pfefferbaum, Ford, Weller, and Kopell (1985) observed the N200 amplitude enhancement in no-go trials using overt behavioral responses and covert cognitive responses. This result is of special importance because it rules out the possibility that the amplitude enhancement of the N200 in no-go trials is due to the absence of motor responses in no-go versus go trials. The N200 enhancement in no-go trials does not seem to be confined to the necessity of withholding motor responses but may appear whenever either overt or covert response preparation has to be interrupted.

De Jong, Coles, Logan, and Gratton (1990) used the stop-signal task to study response inhibition. The N200 was reduced in stop-signal trials where responses were released compared with stop-signal trials where responses were successfully inhibited.¹ Using the flanker task, Gehring, Gratton, Coles, and Donchin (1992) dissociated a central component (200–300-ms poststimulus) and a frontal component (300–450-ms poststimulus) of the N200. The central component was sensitive to the identity of the

stimuli in the visual field irrespective of their spatial location, that is, their status as target or flanker stimuli. The frontal component was specifically sensitive to the flanker compatibility manipulation. The sensitivity of the frontal N200 to the compatibility of the flanker stimuli may be associated with the necessity to interrupt the activation of incorrect responses when competitive responses are primed.

The Present Experiment

To investigate executive control, the flanker task was modified in several respects. Instead of letters with random response assignment, arrowheads were used. The relation between an arrow pointing left and a left-hand response is considered a population stereotype that does not require learning (Kornblum, Hasbroucq, & Osman, 1990). To examine facilitative and interfering flanker effects separately, a neutral flanker condition was included. Several studies showed that facilitation and interference were more pronounced when the onset of the flanker stimuli preceded the target stimulus onset by short time intervals (Eriksen & Schultz, 1979; Flowers, 1990; Flowers & Wilcox, 1982; Grice, Boroughs, & Canham, 1984; Grice & Gwynne, 1985; Taylor, 1977). A constant stimulus onset asynchrony (SOA) of 100 ms was introduced to maximize the expected flanker compatibility effects. Because the impact of flanker stimuli is a function of the spatial distance between flanker and target stimuli (Eriksen & Eriksen, 1974), spatial distance was varied to manipulate the strength of response priming.

In addition to these design modifications, analog response devices instead of response keys were used for assessing partial response activation. The output of these force keys (Giray & Ulrich, 1993) can be used to assess the presence and temporal characteristics of overt responses even if they do not achieve the criterion force level. These partial responses may occur if response activation is insufficiently strong or if the response is aborted before its complete execution. These devices thus allow monitoring of response priming over time. In addition, the electroencephalogram (EEG) can be aligned to various behavioral events that occur below the criterion force level. For example, the EEG can be synchronized to the point in time at which incorrect or correct responses begin to emerge.

Response priming using this task and method should be demonstrable by LRPs and response forces. The resolution of the conflict between responses in the incongruent compatibility condition of the flanker task may follow different routes. The activation of incorrect responses may be insufficiently strong and may wax and wane without any intervention of executive control (Coles, Gehring, Gratton, & Donchin, 1992). Alternatively, the avoidance of incorrect responses may involve active interruption. Clearly, the N200 is related to the interruption of erroneously primed responses in various paradigms. If the N200 is related to the interruption of incorrect responses, then this component should be prominent in the incongruent flanker condition. In addition, the exact nature of the resolution of response conflicts remains to be delineated.

First, the N200 in the incongruent flanker condition may reflect a process that gates the access to effector resources. According to this effector gating model, response conflicts would occur as a consequence of imperfect gating. Low levels of gating would cause high degrees of error, whereas high levels of gating would cause low degrees of error. In this model, the magnitude of erroneous responses should be inversely related to the N200 amplitude.

¹ De Jong, Coles, Logan, and Gratton (1990) did not comment on this N200 amplitude effect. However, inspection of their Figure 5 (p. 176) reveals that the ERPs were more negative during the interval between 300 and 500 ms after the onset of the imperative stimulus in signal-inhibit compared with signal-respond trials.

Second, the appearance of the N200 may reflect inhibitory activity to counter erroneous responses that were automatically primed (Coles, Gratton, & Donchin, 1988). Once the incorrectness of these automatically primed responses has been detected, response priming may be interrupted. According to this response inhibition model, executive control would be exerted by inhibiting inappropriate responses after they were automatically primed and their incorrectness had been discovered. Low degrees of error would require low levels of response inhibition, and high degrees of error would require high levels of response inhibition. Thus, the amplitude of the N200 should covary with the magnitude of the erroneous responses. These predictions can be tested because the force keys allow the assessment of the magnitudes of erroneous responses in single trials. The effector gating model predicts an inverse relationship between degree of error and N200 amplitudes; the response inhibition model predicts a positive relationship between the two measures.

Methods

Subjects

Eighteen paid volunteers served as subjects ($M = 33$ years old; 13 men and 5 women). All subjects were right-handed and all had normal or corrected-to-normal vision.

Stimuli and Procedure

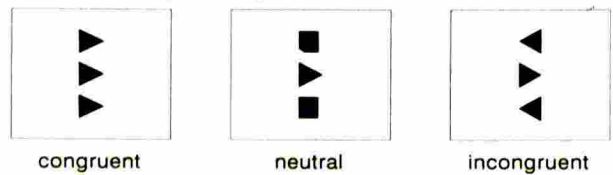
The target stimulus was an arrowhead pointing left or right, presented onto the center of a PC-CRT screen.² Subjects had to respond with the corresponding hand. In congruent trials, two arrows pointing in the same direction as the target stimulus were presented above and below the target. In incongruent trials, two arrows pointing in the direction opposite the target stimulus were presented above and below the target. Squares were used as neutral flanker stimuli in neutral trials. The arrowheads were constructed as equilateral triangles with an edge length of 21 mm; the squares had an edge length of 15 mm. Triangles and squares contained identical numbers of pixels. All stimuli were presented in gray color against a black background. The edge-to-edge distance of flanker and target stimuli was either 1° (short distance) or 3° (large distance) of visual angle (cf. Figure 1).

The beginning of each trial was marked by a central fixation cross presented for 500 ms before the onset of the flanker stimuli. The target stimulus appeared at the center of the CRT screen in addition to the flanker stimuli for 100 ms after presentation of the flanker stimuli. It remained on the screen for 50 ms. All three stimuli were simultaneously removed. A variable intertrial interval of about 2 s followed.

Subjects were seated 1.20 m in front of the CRT screen. They kept their index fingers resting on the response keys. After a practice block of 60 trials, 12 blocks consisting of 60 trials each followed. The data from the practice block were excluded from further analysis. Thus, at least 720 trials were administered during a 45–60-min period. In every block, flanker compatibility (congruent, neutral, incongruent) and spatial distance (short, large) were presented randomly with equal probability. Subjects were instructed to focus on the central target stimulus and to ignore the flanker stimuli. They were instructed to respond as fast as possible at the expense of accuracy.

²For this task, the Experimental Run Time System (ERTS) was used (see Beringer, 1994).

Flanker compatibility



Spatial distance

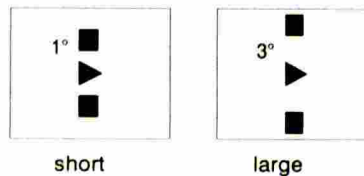


Figure 1. The design of the experiment. Depicted are the three flanker compatibility conditions (congruent, neutral, incongruent) and the two spatial distance conditions (short, large). These various conditions occurred with equal frequencies.

Overt Responses

The motor responses were assessed by means of two analog devices (force keys) that resembled telegraph keys. Each key had a leaf spring that was held by an adjustable clamp at one end; the other end remained mobile. Strain gauges were attached near the fixed end of the leaf spring. Any force applied to the leaf spring led to a displacement at its free end. The displacement was converted into an analogous electrical signal. The electrical signals from both force keys were fed into a second PC, digitized at 100 Hz, and stored on disk. The resolution of this device was about 2 mN. Subjects were asked to rest their fingers on both response keys at all times and to make a response by depressing the appropriate key. They did not exert force on the leaf springs during the interval between successive responses. A response was defined as a key displacement that was equivalent to a force of at least 2 N. The responses were evaluated online. RTs were defined as the interval between the target stimulus onset and the time when a force level of 2 N was achieved with the appropriate key. Subjects were told that they could correct wrong responses by pressing the correct key immediately after the wrong key (i.e., within 1,300 ms after the onset of the flanker stimuli). Uncorrected wrong responses elicited an auditory feedback tone (500 Hz) that began 1,300 ms after the onset of the flanker stimuli. Trials that contained uncorrected errors were repeated at the end of each block of trials. Uncorrected errors occurred very rarely and were not analyzed further. The probabilities of uncorrected errors in the short (large) distance, congruent/neutral/incongruent flanker compatibility condition amounted to 0.09% (0.05%)/0.18% (0.05%)/0.72% (0.23%). Responses that occurred too early (<100 ms) or too late ($>1,200$ ms) and responses that did not achieve the criterion force level of 2 N triggered an alarm tone of 1000 Hz.

Recording Procedure

The EEG was recorded with Ag/AgCl electrodes from the midline sites Fz, Cz, and Pz according to the 10-20 system. The EEG was additionally recorded from C3' and C4' (ca. 1 cm anterior and 1 cm superior to C3 and C4). The EEG recordings were referenced to the left ear lobe. Vertical and horizontal electro-oculograms (EOGs) were recorded from sites above and below

the left eye and 1 cm external to the outer canthus of each eye. A ground electrode was placed on the forehead. The signals were amplified with a Nihon Kohden amplifier (time constant: 10 s; high-frequency cut-off: 35 Hz). Electrode impedance was kept below 5 k Ω . The signals were digitized online at a rate of 100 Hz for 1,400 ms, starting 100 ms before the presentation of the flanker stimuli. No digital filtering was applied to the signals.

Conventional Data Analyses

Stimulus-synchronized average ERPs were computed from correct response trials. This definition excludes the trials in which a preceding wrong response occurred that surpassed 2 N (error corrections). A prestimulus period of 100 ms was subtracted as baseline. Trials containing horizontal eye movements were rejected; trials containing blinks were corrected using the regression method described by Berg (1986). Separate averages were computed (Flanker Compatibility [congruent, neutral, incongruent] $\times$ Spatial Distance [short, large]) for the midline electrode sites (Fz, Cz, Pz). Mean amplitudes for three latency windows (their boundaries were determined by visual inspection of the waveforms) were computed (225–325-ms postflanker: N2b; 250–350-ms posttarget: N2c; 350–450-ms posttarget: P300). The taxonomy for the N200 follows the classification scheme of Pritchard, Shappell, and Brandt (1991). Within their scheme, N2b is related to the attentional detection of deviation from prevailing context and N2c is related to the attentional classification of task relevant stimuli. The latencies and amplitudes of the largest negative peak, 250–350-ms posttarget (N2c peak amplitudes), were computed for the incongruent flanker condition; boundary values were accepted by peak measurement. The LRP was computed from the ERP waveforms recorded at C3' and C4'. The recordings at C3' and C4' were subtracted from one another (left-hand responses: C4' – C3'; right-hand responses: C3' – C4') and averaged separately for left-hand and right-hand responses. These averages were then combined across hands: (left-hand [C4' – C3'] + right-hand [C3' – C4'])/2. Mean amplitudes in an early window (0–100-ms posttarget) and in a late window (150–250-ms posttarget) were computed. The latencies and amplitudes of the largest positive peak in the late window were computed for the incongruent flanker condition.

Graded Response Analyses

Above all, these analyses aimed at the relationship between the resolution of response conflicts and the N200 amplitude. A conflict between flanker- and target-triggered responses was induced specifically in the incongruent flanker condition. The following analyses were thus restricted to this condition. The maximum displacement of the incorrect force key during the RT interval was evaluated for all trials, including error corrections, of the incongruent flanker condition. As a result of the peak measurement, the trials were grouped into three bins of weak, medium, or strong displacement of the incorrect force key (graded response analysis; cf. Coles et al., 1985). One-third of all trials fell into each group. The grouping of trials into force bins was performed within blocks of trials (1, ..., 12), separately for hands (left, right) and separately for spatial distances (short, large). In addition, the onset time of the erroneous response was determined. The onset was defined as the first point in time at which the displacement of the incorrect force key exceeded an arbitrarily chosen onset criterion of 70 mN. Note that the time of onset was determined by a backward searching algorithm that was applied to all trials from the incongruent flanker condition.

First, the maximum displacement of the incorrect force key was determined; all trials of the incongruent flanker condition had some error activity exceeding 70 mN. Second, a backward search along the leading slope of the response deflection was conducted until the first sample point with an amplitude below the pre-defined onset criterion was encountered. The algorithm thus first picked the peak displacement of the incorrect force key and then searched for the point in time when the response deflection fell below the arbitrary onset criterion.

The peak amplitude of the positive deflection of the LRP during the interval between 150 and 300 ms posttarget was computed separately for the three force bins. Stimulus-synchronized average ERPs at the midline electrode sites (Fz, Cz, Pz) for the incongruent flanker condition were averaged separately for the three force bins. The peak-to-peak amplitudes between the largest positive peak 100–300-ms posttarget, the largest negative peak 200–400-ms posttarget, and the baseline-to-peak amplitudes of the largest negative peak 200–400-ms posttarget were computed. These boundaries were again determined by visual inspection of the waveforms. For further analyses, the EEG at the midline electrode sites (Fz, Cz, Pz) was synchronized to the correct and incorrect responses. Positive peaks were picked up within the latency window between –100 and 100 ms relative to the onset of the incorrect responses or the criterion RT of the correct responses. Negative peaks were picked up within the latency window between 0 and 200 ms relative to the onset of the incorrect responses or the criterion RT of the correct responses.

Conservative (Greenhouse–Geisser) *F* tests were used throughout, but original degrees of freedom are reported. Unless indicated, only significant *F* values are presented ($p < .01$). Throughout this paper, the planned comparisons between neutral and congruent and between neutral and incongruent flanker conditions are termed *facilitation* and *interference*, respectively. Post hoc comparisons were made for significant main effects of electrode sites and force bins and for all significant interactions using Bonferroni significance levels to protect against Type I errors.

Results

Conventional Data Analyses

Behavioral findings. A two-way repeated-measures analysis of variance (ANOVA) (flanker compatibility, spatial distance) for mean RTs revealed significant flanker compatibility ($F[2,34] = 722.6$, $\epsilon = 0.837$) and Compatibility $\times$ Distance effects ($F[2,34] = 89.1$, $\epsilon = 0.951$). Facilitation and interference proved significant ($F[1,17] = 378.4$ and 557.7). Spatial distance attenuated both facilitation and interference ($F[1,17] = 52.2$ and 48.1). The ANOVA for error scores revealed significant flanker compatibility ($F[2,34] = 16.2$, $\epsilon = 0.516$), spatial distance ($F[1,17] = 44.4$), and Compatibility $\times$ Distance effects ($F[2,34] = 34.1$, $\epsilon = 0.651$). Facilitation was not apparent in the error scores, but interference was ($F[1,17] = 3.8$, $p = .07$; $F[1,17] = 19.2$); spatial distance reduced interference ($F[1,17] = 30.8$; cf. Table 1).

LRP findings. In Figure 2, the grand mean LRPs are plotted separately for the flanker compatibility (congruent, neutral, incongruent) and spatial distance (short, large) conditions. Incorrect response priming should be reflected by a positive deflection; correct response priming should be reflected by a negative deflection of the LRP. The early LRP deflections (beginning

Table 1. Mean RTs (ms) and Error Scores (%) as a Function of Flanker Compatibility and Spatial Distance

Compatibility	Distance	
	Short	Large
Congruent	306 (2.0)	321 (1.8)
Neutral	350 (5.6)	351 (3.5)
Incongruent	415 (27.0)	395 (15.0)

at around 100-ms posttarget) were positive in the incongruent flanker condition and negative in the congruent flanker condition. Thus, these early deflections indicated that flanker-triggered responses were activated. As late as around 280-ms posttarget, the positive LRP deflection in the incongruent flanker condition was reversed toward a negative LRP deflection. The two-way repeated-measures ANOVA (flanker compatibility, spatial distance) revealed that the mean LRP amplitudes in the early window were unrelated to flanker compatibility, spatial distance, and Compatibility $\times$ Distance ($F_s < 1$). In contrast, the mean amplitudes in the late window were significantly influenced by flanker compatibility ($M = -1.96$ [congruent], -0.24 [neutral], $+1.22$ [incongruent] μV ; $F[2,34] = 67.6$, $\epsilon = 0.585$) and Compatibility $\times$ Distance ($F[2,34] = 11.0$, $\epsilon = 0.879$). Facilitation and interference proved significant ($F[1,17] = 47.2$

and 77.9). Interference was influenced by spatial distance ($F[1,17] = 12.7$), but facilitation was not ($F[1,17] = 2.7$, $p = .12$). A repeated-measures ANOVA (spatial distance) revealed that the amplitude of the positive peak in the incongruent flanker condition was significantly influenced by spatial distance ($M = 2.36$ [short] and 1.83 [large] μV ; $F[1,17] = 12.7$; cf. Figure 2).

ERP findings. In Figure 3, the grand mean ERPs are plotted separately for flanker compatibility (congruent, neutral, incongruent), spatial distance (short, large), and electrode site (Fz, Cz, Pz). Despite the introduction of a 100-ms SOA, the sequence of ERP components in the congruent flanker condition closely resembled that reported by Smid et al. (1990) from Pz in a standard flanker task. Several ERP components were elicited. The early negativity (N100) peaked around 150-ms postflanker (maximum at Pz) and was not analyzed further. The N2b appeared in all flanker compatibility conditions (with a fronto-central maximum). As is evident at Fz and Cz, two distinct positive peaks appeared. The first positive peak appeared with a latency of around 350-ms postflanker or 250-ms posttarget in all compatibility conditions. Abrupt visual onsets are known to trigger automatic modes of attentional allocation (Yantis & Jonides, 1990). We suggest that this positive component reflects the spontaneous orienting of attention associated with the onset of the flanker stimuli. Therefore, it will not be further analyzed. Throughout this paper, P300 refers to the second positive peak. Inspection of Figure 3 suggested that the P300 was reduced in

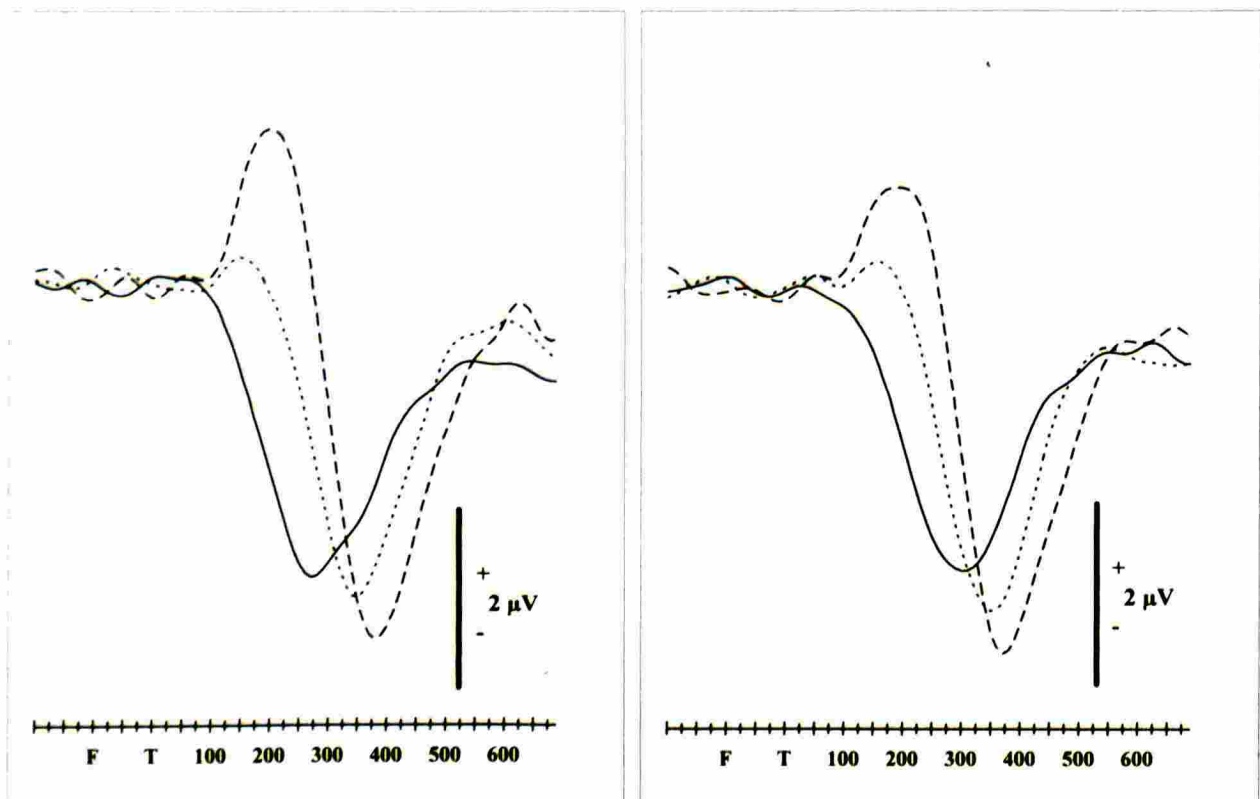


Figure 2. LRPs, averaged for flanker compatibility (congruent: solid line; neutral: dotted line; incongruent: dashed line) and spatial distance (short: left panel; large: right panel). F denotes the onset of flanker stimuli; T denotes the onset of target stimuli. The average waveforms were digitally filtered (low-pass cut-off frequency: 12 Hz).

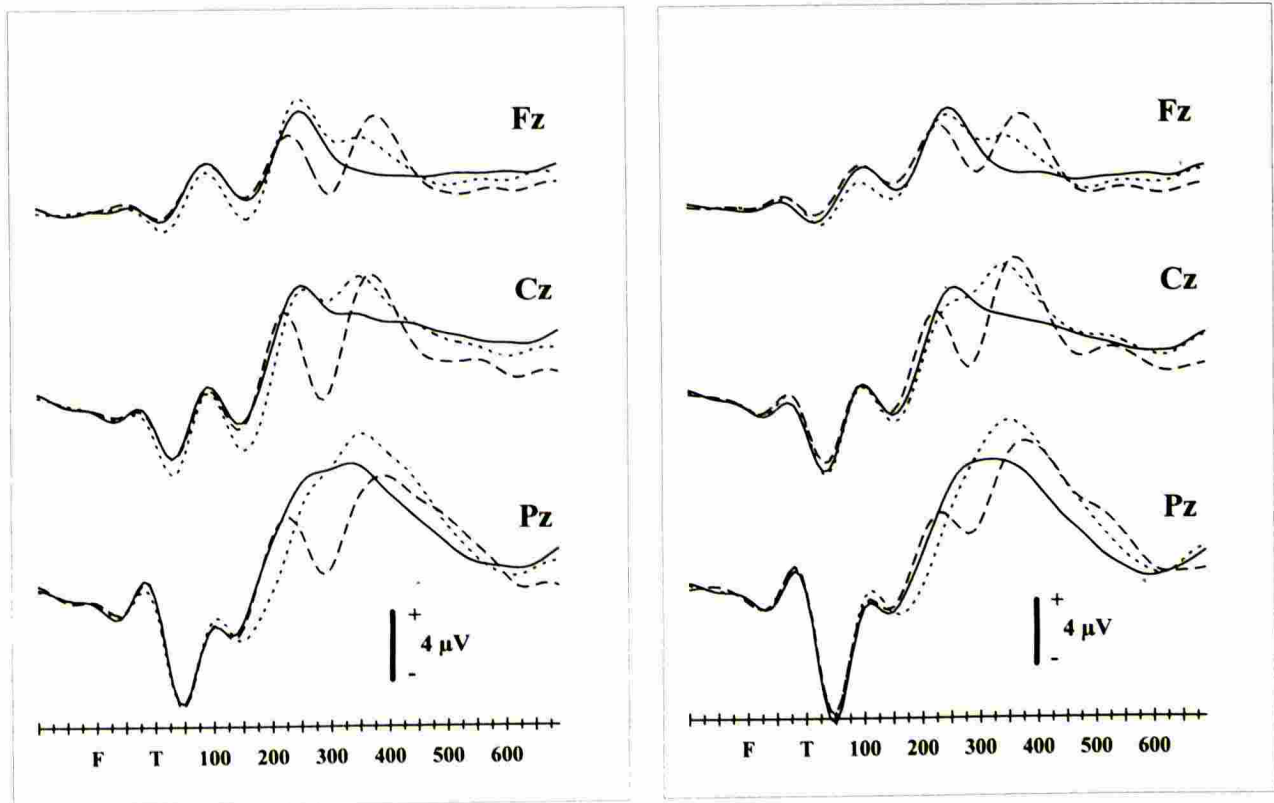


Figure 3. Grand mean ERPs at Fz, Cz, Pz were averaged for flanker compatibility (congruent: solid line; neutral: dotted line; incongruent: dashed line) and spatial distance (short: left panel; large: right panel). F denotes the onset of flanker stimuli; T denotes the onset of target stimuli. The average waveforms were digitally filtered (low-pass cut-off frequency: 12 Hz).

the congruent flanker condition. In addition, a pronounced N2c appeared specifically in the incongruent flanker condition.

A three-way repeated-measures ANOVA (electrode site, flanker compatibility, spatial distance) revealed a significant effect of flanker compatibility on the N2b amplitudes ($M = 1.31$ [congruent], 0.33 [neutral], 1.60 [incongruent] μV ; $F[2,34] = 13.3$, $\epsilon = 0.705$). The N2b was less positive in the neutral flanker condition as both facilitation and interference proved significant ($F[1,17] = 11.2$ and 18.0 ; cf. Figure 3). The ANOVA for the N2c amplitudes revealed significant electrode site ($F[2,34] = 6.8$, $\epsilon = 0.967$), flanker compatibility ($M = 6.06$ [congruent], 6.86 [neutral], 3.81 [incongruent] μV ; $F[2,34] = 19.0$, $\epsilon = 0.806$), Electrode Site $\times$ Compatibility ($F[4,68] = 12.0$, $\epsilon = 0.631$), Compatibility $\times$ Distance ($F[2,34] = 7.3$, $\epsilon = 0.850$), and Electrode Site $\times$ Distance effects ($F[2,34] = 12.2$, $\epsilon = 0.775$). The N2c was more negative at Fz and Cz than at Pz ($F[1,17] \geq 11.8$). Facilitation was not significant, but interference was ($F[1,17] = 4.0$, $p = .06$; $F[1,17] = 37.8$). Interference was smaller at Fz ($M = -1.81$ μV) than at Cz ($M = -3.41$ μV) or at Pz ($M = -3.97$ μV , $F[1,17] \geq 16.5$). Interference was more pronounced in the short than in the large spatial distance condition ($M = -3.72$ [short] and -2.38 [large] μV , $F[1,17] = 7.0$, $p = .02$), although this difference was only marginally significant according to the predefined level of significance. The two-way repeated-measures ANOVA (electrode site, spatial distance) for the N2c peak amplitudes in the incongruent flanker condition revealed significant spatial distance ($M = -0.94$ [short] and $+0.91$ [large] μV , $F[1,17] = 11.9$) and Electrode Site $\times$ Distance effects ($F[2,34] =$

7.2 , $\epsilon = .688$). The Electrode Site $\times$ Distance interaction was reducible to the less pronounced spatial distance effect at Fz compared with the effect at Cz ($M = -0.88$ [Fz] and -2.25 [Cz] μV , $F[1,17] = 12.0$). The N2c peak latencies differed among the electrode sites ($F[2,34] = 7.5$, $\epsilon = 0.867$); this was due to the prolonged peak latencies at Fz ($M = 299$ ms) compared with those at Cz ($M = 286$ ms) or Pz ($M = 288$ ms, $F[1,17] \geq 12.1$). The spatial distance effect on the N2c peak latencies was marginally significant ($M = 293$ [short] and 289 [large] ms, $F[1,17] = 7.2$, $p = .02$). In sum, the N2c appeared specifically in the incongruent flanker condition, and both mean and peak N2c amplitudes were larger when the spatial distance of the flanker stimuli was smaller. In turn, the spatial distance of the flanker stimuli determined the strength of incorrect response activation in the incongruent flanker condition as proved by RTs, error scores, and LRPs.

The three-way ANOVA for the P300 amplitudes revealed significant electrode site ($F[2,34] = 25.1$, $\epsilon = 0.977$), Electrode Site $\times$ Compatibility ($F[4,68] = 9.9$, $\epsilon = 0.676$), Electrode Site $\times$ Distance ($F[2,34] = 15.0$, $\epsilon = 0.907$), and Electrode Site $\times$ Distance $\times$ Compatibility effects ($F[4,68] = 6.8$, $\epsilon = 0.582$). The flanker compatibility effect was marginally significant ($M = 4.64$ [congruent], 6.28 [neutral], 6.22 [incongruent] μV ; $F[2,34] = 6.0$, $p < .02$, $\epsilon = .673$). Facilitation was significant, but interference was not ($F[1,17] = 12.4$, $F[1,17] < 1$). Facilitation and interference were less pronounced at Cz than at Pz ($F[1,17] = 12.1$ and $= 19.3$). The P300 amplitudes were less positive at Fz than those at Cz and Pz ($F[1,17] \geq 15.5$). The spatial distance

effect was more pronounced at Fz than at Cz ($F[1,17] = 11.1$) and at Pz than at Cz ($F[1,17] = 9.6$). The spatial distance effect on interference was more pronounced at Pz than at Cz ($F[1,17] = 7.6$; cf. Figure 3). In sum, the P300 amplitudes were largest at centroparietal electrodes. The P300 was reduced in the congruent flanker condition. At Fz, the P300 had its maximum in the incongruent flanker condition; at Pz, the P300 had its maximum in the neutral flanker condition.

Graded response analyses

Stimulus-locked findings. Several averages were generated separately in temporal synchronization to the onset of the stimuli in the incongruent flanker condition separately for the weak, medium, and strong force bins (cf. Figure 4). The upper left trace depicts the time course of the correct and incorrect motor response. The average amplitude of the stimulus-locked erroneous responses reached less than 0.5 N in the weak, less than 1 N in the medium, and more than 2 N in the strong force bin, that is, the response criterion. The stimulus-locked correct response attained around 5 N irrespective of the strength of the erroneous force. The erroneous response emerged around 250 ms after the onset of the flanker stimuli, whereas the correct response appeared around 250 ms after the onset of the target stimulus. The lower left trace of Figure 4 shows the corresponding LRPs. Importantly, the LRPs of all force bins showed large positive deflections that developed shortly after the onset of

the stimuli (ca. 200-ms postflanker or 100-ms posttarget). A repeated-measures ANOVA (force bin) revealed a significant effect of force bin on the peak amplitudes of the positive LRP deflections ($F[2,34] = 13.8$, $\epsilon = .89$). The strong force bin ($M = +3.17 \mu\text{V}$) yielded greater positivity than the medium force bin ($M = +2.29 \mu\text{V}$, $F[1,17] = 12.4$), with no discernible difference between the weak ($M = +2.11 \mu\text{V}$) and the medium force bins. Stimulus-locked ERPs at the midline scalp sites (Fz, Cz, Pz) for the force bins (weak, medium, strong) are plotted in the right panel of Figure 4. Inspection of these ERP waveforms suggested that the N2c amplitudes covary with the magnitude of the erroneous responses. A two-way repeated-measures ANOVA (electrode site, force bin) revealed significant main effects of electrode site ($F[2,34] = 5.7$, $\epsilon = 0.98$) and force bin ($F[2,34] = 10.4$, $\epsilon = 0.769$) for the baseline-to-peak N2c amplitudes. The N2c amplitudes proved more negative at Cz than those at Pz ($F[1,17] = 10.3$). The strong force bin ($M = -3.02 \mu\text{V}$) yielded greater negativity than did the medium force bin ($M = -0.79 \mu\text{V}$, $F[1,17] = 18.2$), with no discernible difference between the weak ($M = -0.39 \mu\text{V}$) and the medium force bins. A two-way repeated-measures ANOVA (electrode site, force bin) revealed significant main effects of electrode site ($F[2,34] = 23.4$, $\epsilon = 0.893$) and force bin ($F[2,34] = 23.3$, $\epsilon = 0.672$) for the peak-to-peak N2c amplitudes. The peak-to-peak amplitudes proved more negative at Cz than those at Fz and Pz ($F[1,17] > 27.7$). Again, the strong force bin ($M = -11.39 \mu\text{V}$) yielded greater negativity than the medium force bin ($M = -7.98 \mu\text{V}$, $F[1,17] =$

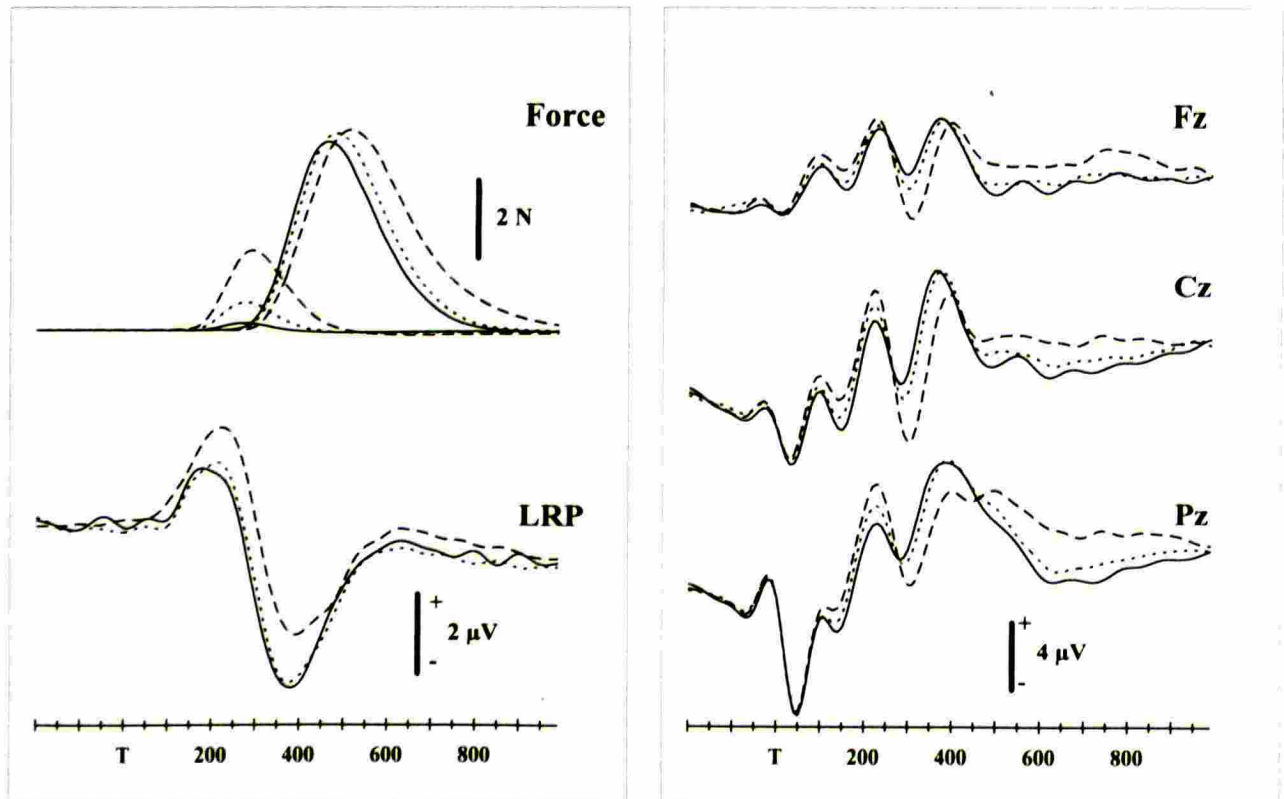


Figure 4. Stimulus-locked ERPs in the incongruent flanker condition broken down by force bins (weak: solid line; medium: dotted line; strong: dashed line). Upper trace of left panel: Correct force: large displacements; incorrect force: small displacements. Lower trace of left panel: LRPs. Right panel: Grand mean ERPs at Fz, Cz, Pz. F denotes the onset of the flanker stimuli; T denotes the onset of the target stimulus. The average waveforms were digitally filtered (low-pass cutoff-frequency: 12 Hz).

18.2), with no discernible difference between the weak ($M = -7.39 \mu\text{V}$) and the medium force bins.

Correct response-locked findings. Next, we aligned the EEG to the point in time at which the correct responses exceeded the criterion force level. Correct response-locked averages are plotted separately for the weak, medium, and strong incorrect force bins in Figure 5. The upper left trace depicts the time course of the correct and incorrect motor responses. The amplitudes of the correct response-locked correct responses amounted to about 6 N. The incorrect responses had already passed their peak amplitudes when the correct responses reached the criterion force level. For example, in the strong force bin, the incorrect responses peaked about 100 ms before the correct responses exceeded the criterion force level. Similar to the stimulus-locked LRPs were the correct response-locked LRPs of all force bins in showing pronounced positivities (cf. lower left trace, Figure 5). Correct response-locked ERPs at the midline scalp sites (Fz, Cz, Pz) for the force bins (weak, medium, strong) are plotted in the right panel of Figure 5. These correct response-locked ERPs differed drastically from the stimulus-locked ERPs (cf. Figure 4). The rapid succession of positive and negative components was replaced by a single positive peak that reached its maximum around the point in time at which the correct responses exceeded the criterion force level. Visual inspection of the correct response-locked ERPs suggested that the peak amplitudes were reduced in the strong force bin compared with the

other force bins. The two-way repeated-measures ANOVA (electrode site, force bin) revealed significant effects of electrode site ($F[2,34] = 9.4$, $\epsilon = .0.972$) and Electrode Site $\times$ Force Bin ($F[4,68] = 5.9$, $\epsilon = .718$) for the peak amplitudes. The main effect of force bin was apparent as a statistical trend ($F[2,34] = 5.0$, $\epsilon = 0.888$, $p < .02$). The positive peak amplitudes proved more positive at Pz than those at Fz ($F[1,17] = 8.7$). Force bin yielded a more pronounced effect at Cz than at Fz ($F[1,17] = 15.9$).

Incorrect response-locked findings. Next, we aligned the EEG to the onset of the incorrect responses.³ Incorrect response-locked averages, broken down by force bins (weak, medium, strong), are plotted in Figure 6. The upper left trace depicts the time course of the correct and incorrect motor responses. This method is charged with an inherent problem: If there is no displacement of the incorrect key in a given trial, then the most pronounced random deviation will be picked up as an incorrect force peak. This false detection of incorrect force peaks induces latency jitter; the frequency of these false detections increases when the mean amplitude of incorrect force decreases. As a consequence, incorrect force bins and the amount of latency jitter

³The particular onset criterion of 70 mN was not crucial to the results. We obtained essentially identical results when we aligned the EEG to the peak displacement of the incorrect force key.

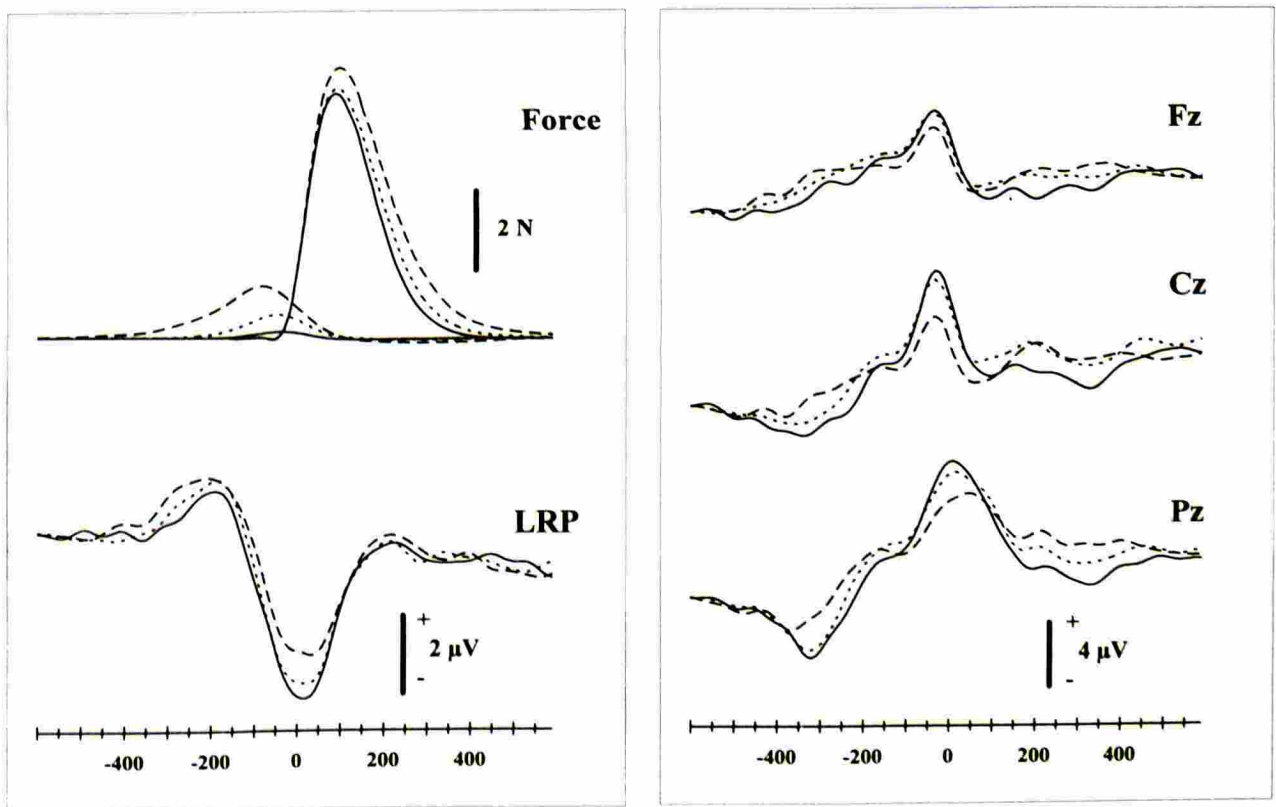


Figure 5. Correct response-locked ERPs in the incongruent flanker condition broken down by force bins (weak: solid line; medium: dotted line; strong: dashed line). Upper trace of left panel: Correct force: large displacements; incorrect force: small displacements. Lower trace of left panel: LRPs. Right panel: Grand mean ERPs at Fz, Cz, Pz. 0 denotes the point in time when the correct response achieved the criterion force level of 2 N. The average waveforms were digitally filtered (low-pass cut-off frequency: 12 Hz).

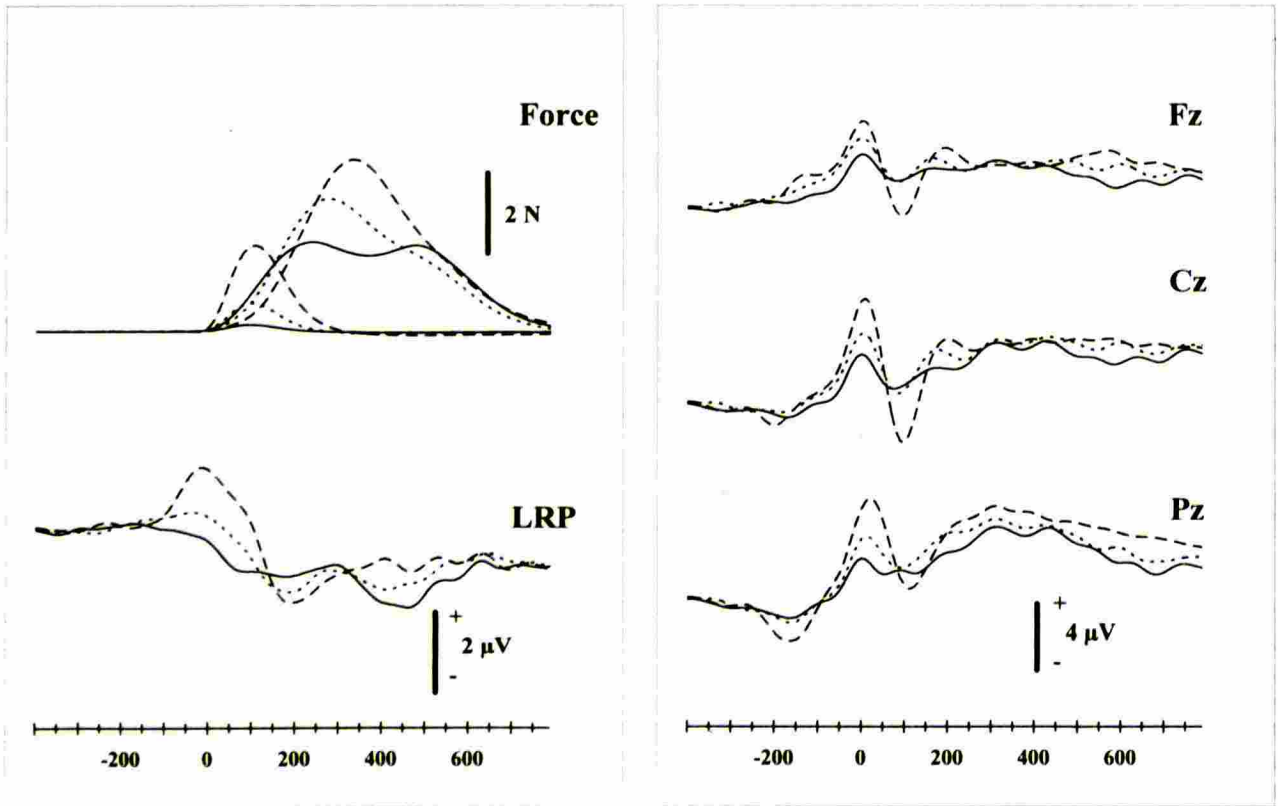


Figure 6. Incorrect response-locked ERPs in the incongruent flanker condition broken down by force bins (weak: solid line; medium: dotted line; strong: dashed line). Upper trace of left panel: Correct force: large displacements; incorrect force: small displacements. Lower trace of left panel: LRPs. Right panel: Grand mean ERPs at Fz, Cz, Pz. 0 denotes the point in time when the incorrect response achieved the criterion force level of 70 mN. The average waveforms were digitally filtered (low-pass cut-off frequency: 12 Hz).

are confounded. The course of the correct responses may serve as a demonstration of the effects of this confounding. The actual course of the correct responses was very similar across force bins (cf. Figure 5). Locked to the onset of incorrect responses, the correct responses appear smeared over the timescale, at least in the lower two force bins (cf. Figure 6).⁴ Thus, the response magnitudes were generally attenuated because of latency jitter, and the degree of that attenuation varied across force bins. As a consequence, we did not compare the force bins directly and restricted our analyses to the strong force bin. Although it is still possible that false peaks were detected in the strong force bin, false detection occurred less frequently in the strong force bin than in the other two force bins.

Visual inspection of these ERPs revealed a rapid succession of a sharp positive and a sharp negative peak. The positive component peaked around the onset of the incorrect responses. It closely resembled the positive peak that was observed in the correct response-locked ERPs. This positive peak may therefore be regarded as an unspecific concomitant of motor responses, be it a correct or an incorrect one. In contrast, the negative deflection appeared in the incorrect response-locked ERPs but not in

the correct response-locked ERPs. Does this signify that this sharp negative deflection is specifically related to the emergence of erroneous responses? To decide this question, in the strong force bin, the negative peaks from the incorrect response-locked ERPs were compared with those from the correct response-locked ERPs. This comparison, however, is not quite fair. First, there is a certain amount of latency jitter in the incorrect response-locked ERPs. Thus, incorrect response-locked ERP amplitudes are attenuated, whereas correct response-locked ERP amplitudes are not. Second, if an ERP component is perfectly synchronized to the emergence of incorrect responses and if there is a temporal dependence between the emergence of incorrect and correct responses, then correct response-locked averages will show an (attenuated) ERP deflection even if there is no direct relationship between the ERP component and the correct response.

The repeated-measures ANOVA (electrode site, response [correct, incorrect]) for the peak-to-peak N2c amplitudes revealed a significant effect of electrode site ($F[2,34] = 19.7$, $\epsilon = 0.847$). The peak-to-peak N2c amplitudes were more pronounced at Cz than at Fz or Pz ($F[1,17] > 16.4$). There was also a marginal Electrode Site $\times$ Response interaction ($F[2,34] = 3.9$, $\epsilon = 0.889$, $p < .04$). This marginal interaction was due to the marginally increased effect of response at Cz compared with that at Fz ($F[1,17] = 6.6$, $p < .02$) or Pz ($F[1,17] = 4.4$, $p < .05$). At single electrodes, the effect of response on the peak-to-peak N2c amplitudes approached significance at the Cz electrode

⁴The distribution of incorrect response onsets was bimodal. Many trials—especially those of the weak force bin—showed very short onset latencies. Absent or low error activity in a given trial obviously increased the chance that enduring activity from the preceding correct response was erroneously picked up as error activity.

($F[1,17] = 2.9, p = .10$). A more clear-cut result emerged when the baseline-to-peak N2c amplitudes were analyzed. The effect of electrode site proved significant ($F[2,34] = 6.5, \epsilon = 0.964$); the effect of response was marginally significant ($F[1,17] = 6.2, p < .03$). The baseline-to-peak N2c amplitudes were more negative at Cz than at Pz ($F[1,17] = 11.9$). There was a significant Electrode Site $\times$ Response interaction ($F[2,34] = 6.9, \epsilon = 0.921$). This interaction was due to the increased effect of response at Cz compared with Fz ($F[1,17] = 17.4$). At single electrodes, the effect of response on the baseline-to-peak N2c amplitudes proved significant at the Cz electrode ($F[1,17] = 8.5$). Despite the fact that the comparison between incorrect response-locked ERPs and correct response-locked ERPs in the strong force bin was biased against this outcome, it became evident that erroneous responses are associated with more pronounced negativities than are correct responses, at least at the Cz electrode.

Discussion

In this flanker task, flanker compatibility exerted large facilitative and interfering effects on performance. Increased spatial distance between target and flanker stimuli attenuated flanker compatibility effects on RTs and error scores. The flanker stimuli induced response priming at cortical and overt levels. Cortical response priming was demonstrable by LRPs; overt response priming was demonstrable by response forces. LRPs proved that flanker-triggered and target-triggered responses were primed at the cortical level in rapid succession. Flanker-triggered LRP deflections were attenuated by the spatial distance of the flanker stimuli. Response forces showed that overt erroneous responses were partially primed in the incongruent flanker compatibility condition. Cortical response priming preceded overt response priming by around 50 ms. Priming of flanker-triggered responses was demonstrable at the cortical level even in the absence of traceable overt erroneous responses in the incongruent flanker compatibility condition. Therefore, response priming has to be viewed as a phenomenon that occurs generally at the cortical level and is not necessarily paralleled by overt response priming. ERP mean amplitudes were influenced by flanker compatibility in several ways: P300 amplitudes were reduced in the congruent flanker condition, N2b amplitudes were enhanced in the neutral flanker condition, and the appearance of a distinct N2c was confined to the incongruent flanker condition. Within the incongruent flanker condition, the magnitude of the erroneous responses covaried with the amplitude of the N2c, which evolved closely synchronized to the emergence of overt erroneous responses. At the vertex electrode, the N2c was specifically related to erroneous responses as compared with motor responses in general.

The P300 was reduced in the congruent flanker condition, that is, when target and flanker stimuli were identical. Identical stimuli allow for response according to any piece of information, irrespective of its flanker or target status; localizing and identifying the target stimulus thus can be omitted. In contrast, when target and flanker stimuli are not identical, as in the neutral and incongruent flanker conditions, additional information processing is necessary (i.e., to localize and identify the target stimulus). The reduced P300 amplitude may reflect the omission of information processing in the congruent flanker condition. The N2b was largest in the neutral flanker condition, but it was not affected differentially by congruent and incongruent flanker stimuli, nor did spatial distance affect this component. Although

the three compatibility conditions were presented with equal probability, flanker stimuli arrowheads were presented in two-thirds of the trials and squares in one-third of the trials. Subjects may thus have perceived arrowheads as the prevailing stimuli and squares as the deviation from the prevailing type of stimulation. If so, the enhanced N2b in the neutral flanker condition reflects the detection of the deviation from the more probable type of stimulation (cf. Pritchard et al., 1991).

Our most important observation was that a distinct N2c was completely confined to the incongruent flanker condition, whereas the neutral and the congruent condition did not differ with respect to the N2c amplitudes. Furthermore, within the incongruent flanker condition, the N2c was larger for flanker stimuli presented close to the target than for flanker stimuli presented farther from the target. The performance, LRP, and response force measures indicated unanimously that the incongruent flanker condition lead to partially primed erroneous responses. We conclude that the N2c appeared when the execution of incorrect responses had to be avoided and was substituted by correct responses. Several other routes of explanation can be excluded. Due to the equiprobable presentation, the flanker compatibility conditions were not confounded with differences in probability, neither does the N2c appear to be related to a perceptual conflict between flanker and target stimuli. According to this view, the N2c should appear whenever flanker and target stimuli are dissimilar. Perceptual conflicts were present in the incongruent and in the neutral flanker conditions, but the emergence of the N2c was confined to the incongruent flanker condition. Our results on the N2b and the N2c converge with those of Gehring et al. (1992) in showing a separation of two N200 subcomponents in the flanker task. The N2b seemed to reflect the detection of a perceptual deviation from a prevailing type of stimulation. The N2c was elicited when erroneous response priming had to be interrupted. Gehring et al. (1992) used a vector filter procedure to disambiguate the two subcomponents of the N200 in the flanker task. In the present investigation, the SOA of 100 ms allowed for dissociation of the N200 subcomponents in the time domain. They were found to be related to the identity of the stimuli in the visual field (N2b) and to the compatibility of the responses that were associated with the flanker and the target stimuli (N2c).

The graded response analysis revealed that the amplitude of the N2c increased as a function of the magnitude of the erroneous responses. In addition, the N2c appeared closely synchronized to the emergence of the incorrect as opposed to the correct responses. In sum, the N2c is sensitive to the avoidance of inappropriate responses. What is the impact of our results on any theory of the neural basis of executive control? First, the mere appearance of the N2c during erroneous response priming indicates that erroneous response priming does not simply wax and wane; it is accompanied by specific neural activity. Second, the positive relationship between erroneous response magnitudes and N2c amplitudes is not compatible with the executive gating model of the interruption of erroneous response priming. This model predicts an inverse relationship between the degree of errors and the N2c amplitudes. This model assumes that response priming occurs subsequent to a gating mechanism such that low levels of erroneous response priming occur as a consequence of high levels of gating. In contrast, the response inhibition model predicts the positive interrelationship between the degree of errors and the N2c amplitudes. According to the response inhibition model, automatically primed erroneous re-

sponses are interrupted once their incorrectness has been detected. Higher levels of erroneous response priming require higher levels of response inhibition. The interruption that is associated with the emergence of the N2c occurred late, that is, after the flanker-triggered responses had already been primed at the cortical level (cf. LRPp). In sum, we suggest that the N2c reflects the inhibition of automatically, but erroneously, primed responses – an executive function. Thus, the N2c clearly differs from the N2pc that emerges with a latency of 200–250 ms at posterior electrode sites when target-similar distractor items are present in visual search tasks (Luck & Hillyard, 1994a). According to Luck and Hillyard (1994b), the N2pc reflects the suppression of competing sensory information that would interfere with the identification of the target. Taken together, the N2c in the flanker task and the N2pc in visual search tasks differ with respect to their functional significance, their latency, and their scalp distribution.

The N2c in the flanker task depends on overt or covert erroneous response priming. Errors in choice RT tasks are associated with more negative ERP amplitudes in the latency range between 300–500 ms after the onset of the imperative stimulus in trials where overt errors occurred compared with correctly responded trials (Falkenstein, Hohnsbein, Hoormann, & Blanke, 1991). This negativity is called *error negativity* (N_E). Because the errors were uncorrected, the initiation of a correction response cannot be regarded as the critical processing component underlying the N_E . N_E had a frontocentral maximum and was larger in incorrect response-triggered averages than in stimulus-triggered averages. Gehring, Goss, Coles, Meyer, and Donchin (1993) analyzed ERPs that were obtained from (partial) errors in the flanker task. The *error-related negativity* (ERN) began around the time of the incorrect response and peaked about 100 ms later. The ERN amplitude was larger when response accuracy was stressed, whereas the ERN amplitude was smaller when the subjects were encouraged to respond quickly. Larger ERN amplitudes were associated with weaker forces with which the incorrect responses were executed, with greater probabilities that an incorrect response was followed immediately by a correct response, and with slower RTs on correct trials following error trials. Because the morphology, the latency, and the scalp distribution of the N2c were very similar to those described for the N_E /ERN (Dehaene, Posner, & Tucker, 1994), we infer that these deflections are similarly related to error processing. Although the N2c and the N_E /ERN seem to be equivalent with respect to their underlying processes, it could be argued that the N_E /ERN reflects unspecifically the activation of attentional processes due to the low frequency of errors and/or the usual consequences of errors (feedback) in choice RT tasks. In contrast, frequency and/or consequence of errors are surely not critical for eliciting the N2c as can be seen easily in our conventional data analyses. These analyses indicate that the study of error processing does not presuppose the presence of complete (uncorrected or corrected) or partial (overt or EMG-defined) errors. For example, the amplitude of the N2c in the incongruent flanker condition was sensitive to the spatial distance of the flanker stimuli that in turn affected the strength of the incorrectly primed responses. We thus infer that the resolution of response conflicts is equivalent to error processing. Inducing response conflicts and manipulating variables that influence the degree of response conflicts thus constitutes an easily applicable neurobehavioral method for studying executive control. We further propose that a particularly valuable neurobehavioral method for investigat-

ing executive control may be the design of response priming studies that use short SOAs between (competitive) distractor and target information in combination with the analysis of the N2c.

The N2c was sensitive to one aspect of executive control, the avoidance of inappropriate action and the selection of appropriate action. We suggest that the N2c is related to the interruption of partially primed responses if their execution leads to an error. Does the N2c reflect error detection or error correction? Let us first assume that N2c reflects the detection of an error, which is a necessary prerequisite for the interruption of inappropriate responses. Little is known about error detection, and several models are conceivable. First, the identification of a mismatch between flanker- and target-triggered motor programs (e.g., Kornblum et al., 1990) may elicit the detection of an error. Target-triggered responses were primed after flanker-triggered responses because we used an SOA of 100 ms. If the N2c is based on the comparison between the two motor programs, it should appear synchronized to the target-triggered responses. This was clearly not the case. Second, exteroceptive or proprioceptive information about the erroneous response may be necessary for error detection. The N2c should emerge synchronized to the errors but with a delay that reflects the time needed for the feedback information to be effective. Proprioceptive feedback is known to be faster than exteroceptive feedback. Nevertheless, the first 100 ms of any movement are considered not to be under the control of proprioceptive information (Jeannerod, 1988; Poulton, 1981). It is unlikely that the N2c is based on proprioceptive (or exteroceptive) feedback because the erroneous response and the N2c emerged simultaneously. Third, internal feedback about the efference for erroneous responses may speed up error correction. Internal feedback involves an “image” of the efference, that is, efference copies (sometimes called *corollary discharges*) that are thought to accompany movements (Taub, 1977). Once the incorrectness of the efference is detected, the execution of the movement could be interrupted immediately. Internal monitoring of the efference would allow very rapid error detection. If error detection is based on internal monitoring, the N2c should appear in advance of any muscular activity that is related to the movement. EMG activity precedes the onset of the movement by around 50 ms (Jeannerod, 1988). The internal feedback model of error detection would be proved if the N2c emerges before any muscular activity takes place. This sequence of events would exclude without doubt a possible role of exteroceptive or proprioceptive feedback. This strong inference is not possible because we did not observe this sequence of events. We suggest that the descending efferent information was internally monitored because the N2c had its onset simultaneously with the erroneous movement. According to this suggestion, error detection involves a comparison between the internally monitored efferent information (what is done) with the accumulated information about the target stimulus (what should be done). However, we base this suggestion on indirect arguments. First, the first 100 ms of movements are not under the control of exteroceptive or proprioceptive feedback. Second, the most popular notion to explain error corrections is a closed-loop model depending on sensory feedback (Schmidt, 1988). However, several studies have shown that rapid error corrections occur in situations that preclude the use of exteroceptive and proprioceptive information (Angel, 1976; Angel, Garland, & Fischler, 1971; Angel & Higgins, 1969; Gibbs, 1965; Higgins & Angel, 1970; Megaw, 1972; Rabbitt, 1966a, 1966b, 1968). In a choice RT task involving a quick joystick movement, Kopp and

Rist (1994) found that around 50% of the errors were corrected; 93.2% of these error corrections were initiated within 90 ms, at most. Third, Diggles (1987) showed that error correction latencies during passive movements ($M = 224$ ms) are considerably prolonged compared with those during active movements ($M = 84$ ms). In passively induced errors, error detection must rely on proprioceptive information because no internal efference signal is available to supply rapid error information as with actively induced errors. Fourth, Cooke and Diggles (1984) showed that the correction of errors is detectable some 50 ms before the actual movement begins, that is, EMG activity is suppressed that otherwise would have produced the error movement. This suppression cannot be based on proprioceptive information. "It is suggested that this earliest response to the error movement is based on central monitoring of the commands for movement" (Cooke & Diggles, 1984, p. 348).

Does the N2c reflect the correction process that (probably) follows the detection of an error? Equally little is known about

the mechanisms of error correction. For example, does the interruption of errors involve inhibitory neural activity? The N200 has been observed with similar latencies and topographies in no-go trials, reflecting the necessity to withhold motor responses. To the extent that the N200 in no-go trials is associated with response inhibition, the emergence of the N2c in the incongruent condition of the flanker task provides a clue that erroneous responses receive inhibition. The generality of cortical response priming implies that the inhibition of inappropriate responses is a general principle of executive control.

Our results do not allow a decision as to whether the N2c reflects the detection or the correction of erroneously primed responses. Importantly, the interruption of inappropriate actions extends to motor and cognitive responses. Therefore, a central executive seems to be an appropriate theoretical level (Dehaene et al., 1994). The regulatory activity of the central executive does not occur specifically in the motor system; this merely offers an opportunity to study executive control precisely.

REFERENCES

- Allport, A. (1989). Visual attention. In M. I. Posner (Ed.), *Foundations of cognitive science* (pp. 631–682). Cambridge, MA: Bradford.
- Angel, R. W. (1976). Efference copy in the control of movement. *Neurology*, 26, 1164–1168.
- Angel, R. W., Garland, H., & Fischler, M. (1971). Tracking errors amended without visual feedback. *Journal of Experimental Psychology*, 89, 422–424.
- Angel, R. W., & Higgins, J. R. (1969). Correction of false moves in pursuit tracking. *Journal of Experimental Psychology*, 82, 185–187.
- Berg, P. (1986). The residual after correcting event-related potentials for blink artifacts. *Psychophysiology*, 23, 354–364.
- Beringer, J. (1994). Software announcements: ERTS: A flexible software tool for developing and running psychological reaction time experiments on IBM PCs. *Behavior Research Methods, Instruments, and Computers*, 55(4), 1.
- Coles, M. G. H. (1989). Modern mind-brain reading: Psychophysiology, physiology, and cognition. *Psychophysiology*, 26, 251–269.
- Coles, M. G. H., Gehring, W. J., Gratton, G., & Donchin, E. (1992). Response activation and verification: A psychophysiological analysis. In G. E. Stelmach & J. Requin (Eds.), *Tutorials in motor behavior II* (pp. 779–792). Amsterdam: Elsevier.
- Coles, M. G. H., Gratton, G., Bashore, T. R., Eriksen, C. W., & Donchin, E. (1985). A psychophysiological investigation of the continuous flow model of human information processing. *Journal of Experimental Psychology: Human Perception and Performance*, 11, 529–553.
- Coles, M. G. H., Gratton, G., & Donchin, E. (1988). Detecting early communication: Using measures of movement-related potentials to illuminate human information processing. *Biological Psychology*, 26, 69–89.
- Cooke, J. D., & Diggles, V. A. (1984). Rapid error correction during human arm movements: Evidence for central monitoring. *Journal of Motor Behavior*, 16, 348–363.
- de Jong, R., Coles, M. G. H., Logan, G. D., & Gratton, G. (1990). In search of the point of no return: The control of response processes. *Journal of Experimental Psychology: Human Perception and Performance*, 14, 164–182.
- Dehaene, S., Posner, M. I., & Tucker, D. M. (1994). Localization of a neural system for error detection and compensation. *Psychological Science*, 5, 303–305.
- Diggles, V. A. (1987). Rapid error corrections: Behavioral evidence for central monitoring of efference. *Journal of Human Movement Studies*, 13, 433–460.
- Eimer, M. (1993). Effects of attention and stimulus probability on ERPs in a go/nogo task. *Biological Psychology*, 35, 123–138.
- Eriksen, B. A., & Eriksen, C. W. (1974). Effects of noise letters upon the identification of a target letter in a nonsearch task. *Perception and Psychophysics*, 16, 143–149.
- Eriksen, C. W., & Schultz, D. W. (1979). Information processing in visual search: A continuous flow conception and experimental results. *Perception and Psychophysics*, 25, 249–263.
- Falkenstein, M., Hohnsbein, J., Hoormann, J., & Blanke, L. (1991). Effects of crossmodal divided attention on late ERP components. II. Error processing in choice reaction tasks. *Electroencephalography and Clinical Neurophysiology*, 78, 447–455.
- Flowers, J. H. (1990). Priming effects in perceptual classification. *Perception & Psychophysics*, 47, 135–148.
- Flowers, J. H., & Wilcox, N. (1982). The effect of flanking context on visual classification: The joint contribution of interactions at different processing levels. *Perception and Psychophysics*, 32, 581–591.
- Gehring, W. J., Goss, B., Coles, M. G. H., Meyer, D. E., & Donchin, E. (1993). A neural system for error detection and compensation. *Psychological Science*, 4, 385–390.
- Gehring, W. J., Gratton, G., Coles, M. G. H., & Donchin, E. (1992). Probability effects on stimulus evaluation and response processes. *Journal of Experimental Psychology: Human Perception and Performance*, 18, 198–216.
- Gibbs, C. B. (1965). Probability learning in step-input tracking. *British Journal of Psychology*, 56, 233–242.
- Giray, M., & Ulrich, R. (1993). Motor coactivation revealed by response force in divided and focused attention. *Journal of Experimental Psychology: Human Perception and Performance*, 19, 1278–1291.
- Gratton, G., Coles, M. G. H., Sirevaag, E. J., Eriksen, C. W., & Donchin, E. (1988). Pre- and poststimulus activation of response channels: A psychophysiological analysis. *Journal of Experimental Psychology: Human Perception and Performance*, 14, 331–344.
- Grice, G. R., Boroughs, J. M., & Canham, L. (1984). Temporal dynamics of associative interference and facilitation produced by visual context. *Perception and Psychophysics*, 36, 499–507.
- Grice, G. R., & Gwynne, J. W. (1985). Temporal characteristics of noise conditions producing facilitation and interference. *Perception and Psychophysics*, 37, 495–501.
- Higgins, J. R., & Angel, R. W. (1970). Correction of tracking errors without sensory feedback. *Journal of Experimental Psychology*, 84, 412–416.
- Jeannerod, M. (1988). *The neural and behavioural organization of goal-directed movements*. Oxford: Clarendon.
- Jodo, E., & Kayama, Y. (1992). Relation of a negative ERP component to response inhibition in a go/no-go task. *Electroencephalography and Clinical Neurophysiology*, 82, 477–482.
- Kok, A. (1986). Effects of degradation of visual stimulus components of the event-related potential (ERP) in go/nogo reaction tasks. *Biological Psychology*, 23, 21–38.
- Kopp, B., & Rist, F. (1994). Error-correcting behavior in schizophrenic patients. *Schizophrenia Research*, 13, 11–22.
- Kornblum, S., Hasbroucq, T., & Osman, A. (1990). Dimensional overlap: Cognitive basis for stimulus-response compatibility – A model and taxonomy. *Psychological Review*, 97, 253–270.

- Kutas, M., & Donchin, E. (1980). Preparation to respond as manifested by movement-related brain potentials. *Brain Research*, 202, 95–115.
- Logan, G. D., & Cowan, W. B. (1984). On the ability to inhibit thought and action: A theory of an act of control. *Psychological Review*, 91, 295–327.
- Luck, S. J., & Hillyard, S. A. (1994a). Electrophysiological correlates of feature analysis during visual search. *Psychophysiology*, 31, 291–308.
- Luck, S. J., & Hillyard, S. A. (1994b). Spatial filtering during visual search: Evidence from human electrophysiology. *Journal of Experimental Psychology: Human Perception and Performance*, 20, 1000–1014.
- Megaw, E. D. (1972). Directional errors and their correction in a discrete tracking task. *Ergonomics*, 15, 633–643.
- Neumann, O. (1990). Visual attention and action. In O. Neumann & W. Prinz (Eds.), *Relationships between perception and action. Current approaches* (pp. 227–267). Berlin: Springer.
- Pfefferbaum, A., & Ford, J. M. (1988). ERPs to stimuli requiring response production and inhibition: Effects of age, probability and visual noise. *Electroencephalography and Clinical Neurophysiology*, 71, 55–63.
- Pfefferbaum, A., Ford, J. M., Weller, B. J., & Kopell, B. S. (1985). ERPs to response production and inhibition. *Electroencephalography and Clinical Neurophysiology*, 60, 423–434.
- Poulton, E. C. (1981). Human manual control. In V. B. Brooks (Ed.), *Handbook of physiology: Sec. 1. The nervous system: Vol. 2. Motor control, Part 2* (pp. 1337–1389). Bethesda, MD: American Physiological Society.
- Pritchard, W. S., Shappell, S. A., & Brandt, M. E. (1991). Psychophysiology of N200/N400: A review and classification scheme. *Advances in Psychophysiology*, 4, 43–106.
- Rabbitt, P. M. A. (1966a). Error correction time without external error signals. *Nature*, 212, 438.
- Rabbitt, P. M. A. (1966b). Errors and error correction in choice-response tasks. *Journal of Experimental Psychology*, 71, 264–272.
- Rabbitt, P. M. A. (1968). Three kinds of error-signalling responses in a serial choice task. *Quarterly Journal of Experimental Psychology*, 20, 179–188.
- Schmidt, R. A. (1988). *Motor control and learning. A behavioral emphasis*. Champaign, IL: Human Kinetics Publishers.
- Schröger, E. (1993). Event-related potentials to auditory stimuli following transients shifts of spatial attention in a go/nogo task. *Biological Psychology*, 36, 183–207.
- Smid, H. G. O. M., Mulder, G., & Mulder, L. J. M. (1990). Selective response activation can begin before stimulus recognition is complete: A psychophysiological and error analysis of continuous flow. *Acta Psychologica*, 74, 169–201.
- St. James, J. D. (1990). Observations in the microstructure of response conflict. *Perception and Psychophysics*, 48, 517–524.
- Taub, E. (1977). Movement in nonhuman primates deprived of somatosensory feedback. In J. F. Keogh (Ed.), *Exercise and sports sciences reviews* (Vol. 4, pp. 335–374). Santa Barbara: Journal Publishing Affiliates.
- Taylor, D. A. (1977). Time course of context effects. *Journal of Experimental Psychology: General*, 106, 404–426.
- Yantis, S., & Jonides, J. (1990). Abrupt visual onsets and selective attention: Voluntary versus automatic allocation. *Journal of Experimental Psychology: Human Perception and Performance*, 16, 121–134.

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